Cohort-level differential distributional analysis for studying microglia in Alzheimer's disease via single-cell RNA-sequencing

Wenjing Tati Zhang[1], Jinqiu Turbo Du[1], Yihan Chen[1], Suman Jayadev[2], Katherine E. Prater[2], Kevin Z. Lin[1]

1 University of Washington (UW) Biostatistics
2 UW Neurology

Motivation

- Many snRNA-seq datasets and differential expression (DE) methods – what experimental and statistical considerations make findings more reproducible?
- Most DE methods also only focus on differences in mean expression – are there other types of differences that are biologically insightful?

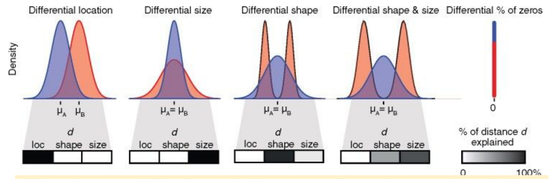

Fig. 1: waddR (Schefzik et al., Bioinformatics, 2021)

Datasets

- Focus on snRNA-seq microglia datasets from post-mortem frozen brain tissue (PFC)
  - **Prater et al. (Nature Aging, 2023)** of 22 donors from UW ADRC
  - **SEA-AD (Gabitto et al., Nature Neuro., 2024)** of 80 donors from ACT and UW UDRC
  - **ROS-MAP (Sun et al., Cell, 2023)** of 345 donors from ROS and MAP

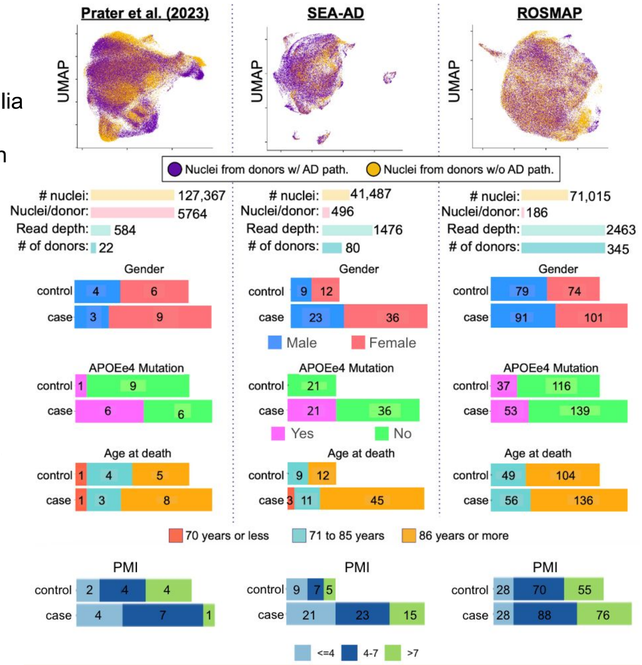

Fig. 2: Overview of the three datasets

Methods

- DE between donors w/ AD pathology ("case") from w/o AD pathology ("control"), where statistical power is dictated by # of donors, not # of cells.
- Focus on pseudobulking-DESeq2[1], eSVD-DE[2], and NEBULA[3], with additional results (not shown here) using pseudobulking-MAST[4] and Dreamlet[5]
- Adjusted for: Age at death, sex, APOE genetics, ethnicity, PMI, sequencing batch

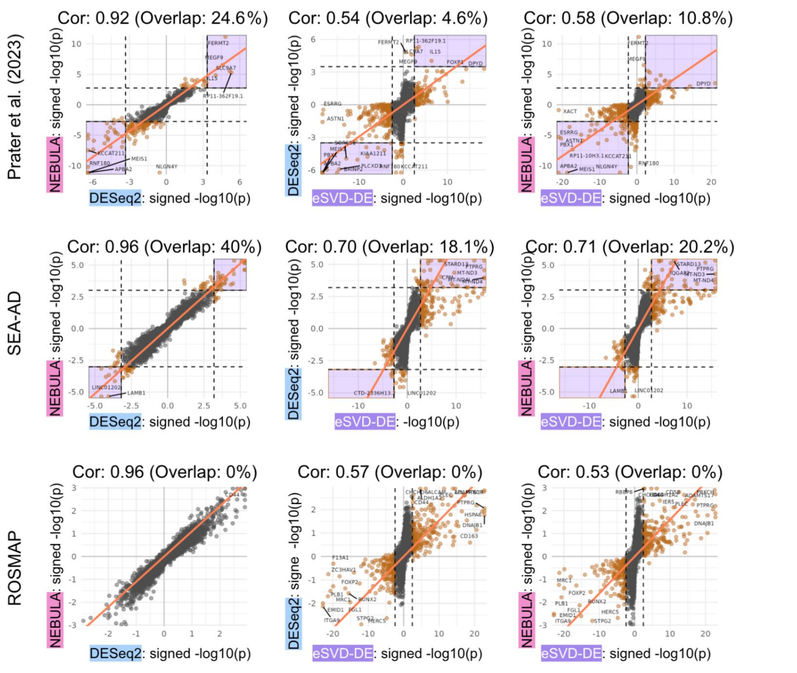

Fig. 3: Initial results. Each point is a gene, signed -log10 p-value for each method, dashed line for multiplie-testing threshold. Orange line: line of correlation. Purple region: DE genes using multiple methods. Correlation: Correlation -log10 p-values

Comparison between methods

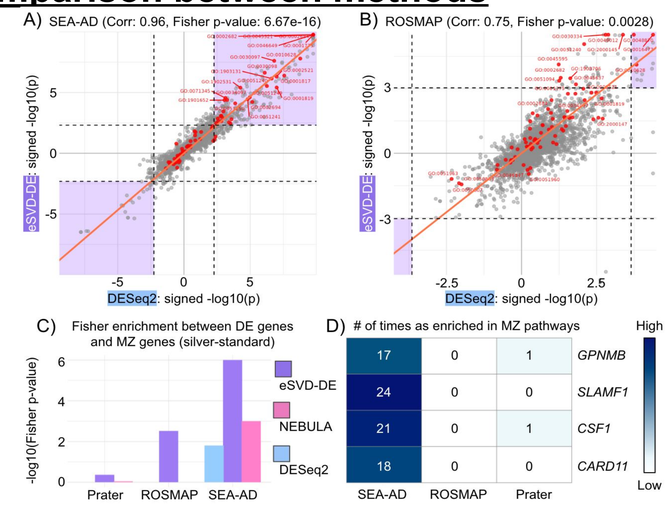

Fig. 4: Comparing 2 DE methods on 1 dataset. A and B) Signed -log10 p-values for GSEA, comparing eSVD-DE or DESeq2, on SEA-AD or ROSMAP. Highlighted pathways are "silver-standard" pathways from MZ[6,7] {Pathways about: immune regulation, cell signaling, motility}. Highest concordance in SEA-AD. C) Enrichment comparing DE methods with silver-standard. D) Number of times key genes are included in enriched pathways.

Comparison between datasets

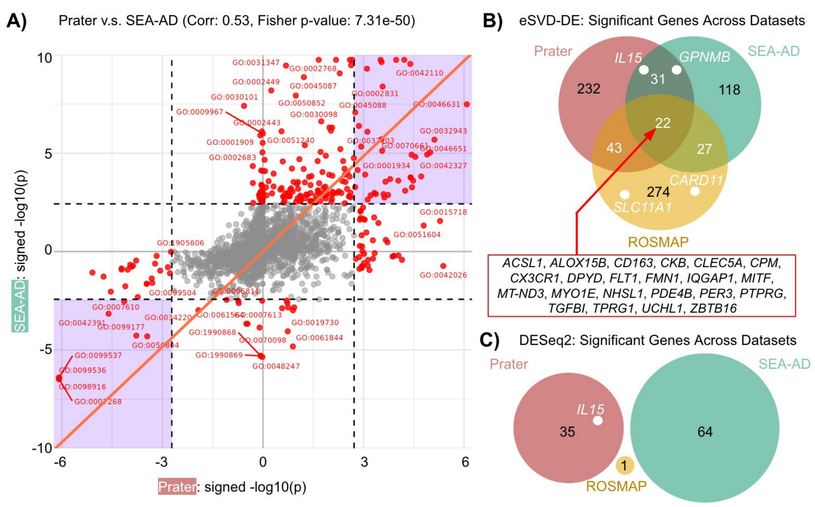

Fig. 5: Comparing 2 datasets analyzed using 1 method. A) Comparison using eSVD-DE to analyze SEA-AD and Prater. One point per pathway. B and C) Overlap in DE genes using eSVD-DE or DESeq2, respectively. MZ genes marked.

Downsampling experiments

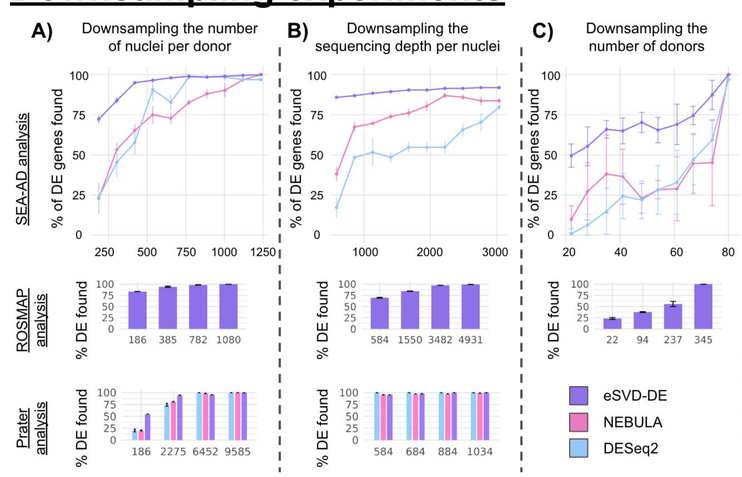

Fig. 6: Downsampling experiment to determine the experimental factor that drives power. A) Downsampling the number of nuclei per donor. B) Downsampling the sequencing depth of each nuclei. C) Downsampling the number of donors, holding case-control and sex balance fixed. The results suggest that the number of the donors dictates the statistical power the most.

Looking beyond differential means

- Mathematically, we use a known decomposition of the *Wasserstein distance* between two distributions, decomposing into differences between the two distributions' mean, standard deviation, or shape.

$$W_2^2(P,Q) = (\mu_P - \mu_Q)^2 + (\sigma_P - \sigma_Q)^2 + 2\sigma_P\sigma_Q\rho_{PQ}$$

We develop a framework: For one gene at a time, we treat each donor's cells as a distribution. We ask: Are there genes whose distributions differ significantly b/w case and control, but not their mean?

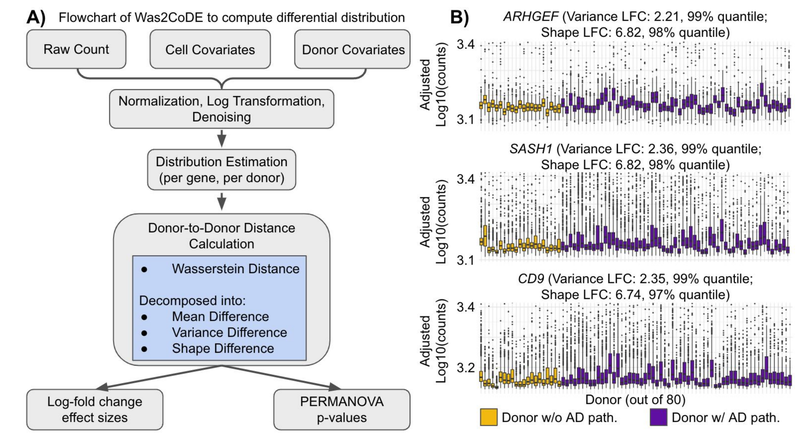

Fig. 7: Was2CoDE (Wasserstein-2 based Cohort Differential Expression). A) Workflow. B) Highlight of microglial genes involved in immune signaling and phagocytosis that have substantial differentiation distribution, but not mean.

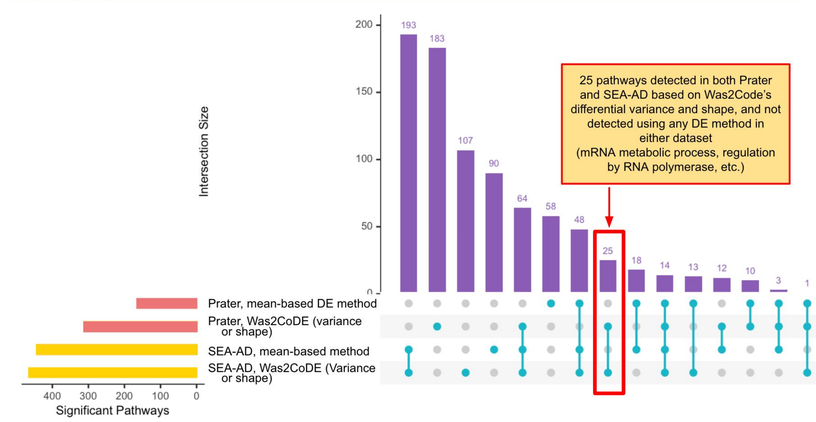

Fig. 8: Pathways uniquely detected by Was2CoDE found in multiple datasets. We find pathways (RNA metabolic processes including regulation of RNA biosynthetic process, mRNA metabolic process, and regulation of transcription by RNA polymerase II) found in both Prater and SEA-AD using Was2CoDE, but not found using DESeq2, eSVD-DE, or NEBULA.

Conclusions

- Cohort design and DE methods seem to be the most influential factors that drive reproducibility
- Mean-centric DE tests overlook variance/shape shifts that may pinpoint actionable immune and motility programs
- While many microglial AD studies use DE methods on the cell level (i.e., simply having more cells yields more power), DE methods on the donor level would yield more biologically relevant findings

Citations: 1) Love et al., Genome Bio., 2014; **2) Lin et al.**, BMC Bioinf., 2024; **3) He et al.**, Com. Bio., 2021; **4) Finak et al.**, Genome Bio., 2015; **5) Hoffman et al.**, bioRxiv, 2024; **6) Mathys et al.**, Nature, 2024; **7) Zhou et al.**, Nature Med., 2020